

Practical 3

Aim: Import a dataset with missing values, inspect the data, and handle missing values using techniques like deletion, mean imputation, and forward/backward fill.

Source-Code:

```
import pandas as pd
import numpy as np

data = {
    "Name": ["Aman", "Riya", "Rahul", "Sneha", "Karan"],
    "Age": [20, 21, np.nan, 22, 23],
    "Marks": [85, np.nan, 78, 90, np.nan]
}
df = pd.DataFrame(data)

print("===== Original Dataset =====\n")
print(df)

print("\n===== Dataset Information =====\n")
print(df.info())

print("\n===== Missing Values =====\n")
print(df.isnull().sum())

deleted_data = df.dropna()

print("\n===== Dataset After Removing Missing Values =====\n")
print(deleted_data)

mean_data = df.copy()

age_mean = mean_data["Age"].mean()

mean_data["Age"] = mean_data["Age"].fillna(age_mean)

marks_mean = mean_data["Marks"].mean()

mean_data["Marks"] = mean_data["Marks"].fillna(marks_mean)

print("\n===== Dataset After Mean Imputation =====\n")
print(mean_data)

forward_fill_data = df.ffill()

print("\n===== Dataset After Forward Fill =====\n")
print(forward_fill_data)

backward_fill_data = df.bfill()

print("\n===== Dataset After Backward Fill =====\n")
print(backward_fill_data)
```

Output:

===== Original Dataset =====

	Name	Age	Marks
0	Aman	20.0	85.0
1	Riya	21.0	NaN
2	Rahul	NaN	78.0
3	Sneha	22.0	90.0
4	Karan	23.0	NaN

===== Dataset Information =====

```
<class 'pandas.DataFrame'>
RangeIndex: 5 entries, 0 to 4
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype
---  -
 0   Name    5 non-null      str
 1   Age     4 non-null      float64
 2   Marks   3 non-null      float64
dtypes: float64(2), str(1)
memory usage: 275.0 bytes
None
```

===== Missing Values =====

```
Name      0
Age        1
Marks      2
dtype: int64
```

===== Dataset After Removing Missing Values =====

	Name	Age	Marks
0	Aman	20.0	85.0
3	Sneha	22.0	90.0

===== Dataset After Mean Imputation =====

	Name	Age	Marks
0	Aman	20.0	85.000000
1	Riya	21.0	84.333333
2	Rahul	21.5	78.000000
3	Sneha	22.0	90.000000
4	Karan	23.0	84.333333

===== Dataset After Forward Fill =====

	Name	Age	Marks
0	Aman	20.0	85.0
1	Riya	21.0	85.0
2	Rahul	21.0	78.0
3	Sneha	22.0	90.0
4	Karan	23.0	90.0

===== Dataset After Backward Fill =====

	Name	Age	Marks
0	Aman	20.0	85.0
1	Riya	21.0	78.0
2	Rahul	22.0	78.0
3	Sneha	22.0	90.0
4	Karan	23.0	NaN